

ASHUTOSH VERMA

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SUMMARY

Senior Software Engineer with 8+ years of experience designing **high-throughput backend systems**, **real-time ETL pipelines**, and **distributed architectures**. Expert in **Python**, **FastAPI**, and **modular API design** with deep expertise in **Test Driven Development (TDD)**, **database schema design**, and **performance optimization**. Specialized in scalable backend development, concurrency, and system reliability with strong fundamentals in **data modeling** and **clean architecture**.

IMPACT HIGHLIGHTS

- Engineered async job orchestration scaling to 20k+ concurrent executions — reduced API latency from 40s → 50ms (p99) and achieved **22× faster ETL pipelines** with PySpark.
- Replatformed monolith to **microservices** with HA clusters and observability, achieving 99.99% uptime and 60% lower MTTR.
- Built **modular APIs** and **AI chatbot (FastAPI + VectorDB + Elasticsearch)** automating 80% of user queries and enabling proactive anomaly detection ([Repo](#)).

EXPERIENCE (8 YEARS)

Fluxon Pvt. Ltd.

Senior Software Engineer (Contract)

Remote, India
Mar 2026 – Present

- Led the architecture and development of an enterprise-wide **AI Governance, Cost Intelligence, and Observability Platform** serving **400+ employees** across **20+ AI platforms**, including Claude, OpenAI, and other leading AI tools.
- Designed scalable backend services using **NestJS, TypeScript, BigQuery, PostgreSQL, GCP, Docker, and CI/CD pipelines** to aggregate, process, and analyze millions of AI usage and telemetry events across teams.
- Built advanced **AI cost analytics and optimization dashboards** with team-wise RBAC, budget controls, consumption forecasting, utilization insights, and configurable usage-alert policies, enabling proactive spend governance across the organization.
- Implemented **GenAI-powered anomaly detection and pattern analysis systems** to identify unusual usage spikes, cost outliers, inefficient prompt patterns, token consumption anomalies, and abnormal platform behavior before budget thresholds were breached.
- Developed intelligent alerting workflows supporting configurable monitoring windows (hourly, daily, weekly, monthly) with automated escalation and recommendation engines for cost optimization and operational governance.
- Integrated **Claude AI** into engineering workflows to automate pull request reviews, code quality validation, security checks, documentation feedback, and fix recommendations, reducing manual review effort by approximately **90%**.
- Established enterprise-grade access controls using **RBAC**, audit logging, usage governance, and team-level visibility, providing leadership with centralized insights into AI adoption, utilization, and ROI metrics.

NAV Backoffice Pvt. Ltd.

Platform Engineer

Jaipur, India
Aug 2024 – Present

- Engineered high-performance **Python/FastAPI services**, automated reporting, and reconciliation workflows integrating with **modular APIs**.
- Detected and fixed a FastAPI memory leak (1.5TB over 5 days) by profiling asyncio loops and cache lifetimes — stabilized infra and cut infra cost by 35%.
- Implemented **TDD pipelines** with 90%+ coverage and scalable database schema design for financial data ingestion.
- Developed a **retrieval-augmented chatbot (RAG)** improving customer query resolution by 70–80%.
- Provided L3 production support and root-cause analysis across microservices — reduced incident MTTR by 35 %.

- Designed multi-tenant ETL pipelines for marketplace analytics using **Python, Cassandra, Airflow, and Kestra** — processed billions of events/day.
- Migrated APIs to **SQLite** with enforced **unit and integration testing (TDD)** — achieved 98% coverage and resolved 90+ critical defects.
- Delivered **scalable service integrations** (pricing engine, POS platform) for 2,600+ U.S. based Stores (Petco).

- Led architecture and development of a high-performance **Next.js PWA** and **React Native** app ([ornaz.com](#)), achieving a [perfect 100/100 Lighthouse score](#).
- Migrated the legacy monolithic system to **cloud-native microservices** with distributed workers and fault-tolerant.
- Built a real-time **AR Try-On** experience ([Demo](#)) using C++, MediaPipe, WebGL, WASM — increased conversion.
- Developed a robust **CRM and API-driven workflow system** (Django + Celery + RabbitMQ).
- Architected containerized infrastructure (**Docker Swarm**) and observability stack (Grafana, ELK, Jaeger) reducing MTTR by 60%.

SELECTED PROJECTS

- Semantic Search Chatbot:** Retrieval-augmented chatbot using **FastAPI, LangChain, Vector DB** — achieved 80% automated query resolution ([Architecture Write-up](#)).
- Accounts API Optimization:** Reduced NAV Accounts API latency from 40s → 50ms (p99) using **async workers, caching, and optimized query planning**.
- Realtime AR Try-On:** Built browser and mobile AR experience using **C++, WebGL, WASM, Bazel** — enabled 30% higher conversions with GPU-accelerated rendering.
- Kestra Cassandra Plugin:** Contributed to open-source connector stability ([PR #45](#)) improving ingestion reliability.

TECHNICAL SKILLS

Languages	Python, Go, JavaScript/TypeScript, C++, C#, SQL, Node.js
Frameworks	FastAPI, Django, React, Next.js, Nest.js, GraphQL
Data & Stream	PostgreSQL, Cassandra, Kafka, Redis, Airflow, Kestra
Testing & TDD	Pytest, UnitTest, Coverage, Mocking, CI Integration
AI/ML	LangChain, OpenAI API, RAG Pipelines , Elasticsearch
DevOps / Infra	Docker, Kubernetes, Terraform, Jenkins, Helm, AWS, GCP
Observability	Grafana, Prometheus, Jaeger, ELK, Sentry
Core Skills	TDD, API Design, Database Modeling, Scalability, System Design, CI/CD

LEADERSHIP & COMMUNITY

- Mentored engineers and enforced **TDD and coding standards** improving delivery quality.
- Spearheaded POS rollout across 2,600+ U.S. retail stores at ShyftLabs.
- Drove performance and code-review culture at NAV, improving release reliability.
- Authored technical blogs and guides on [ashutoshverma.com](#).
- Contributed to open source: **Kestra Cassandra plugin** ([PR #45](#)).
- Earned 3 consecutive “**Technical Excellence Award**” at NAV for leading Payoneer project excellence and engineering best practices.

EDUCATION